

Analysis of Blinking Yellow Signal Detection

Team 1

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1 Objective

The objective of this work was to design and experimentally verify an algorithm for distinguishing between two states of a yellow railway light signal in video footage: a continuously illuminated yellow signal and a blinking yellow signal. The input to the algorithm is a video, or alternatively a CSV file generated by the detection pipeline. The output is the classification of the signal state together with diagnostic outputs.

2 Method

The processing of a single video is divided into several steps:

1. detection of the object of type `signal` using a YOLO model,
2. saving detections into a CSV file,
3. selection of the tracked `detection_ID`,
4. extraction of the `bbox` region from the video,
5. computation of color masks for the yellow and overexposed parts of the signal,
6. creation of a brightness signal over time,
7. removal of the slow trend,
8. detection of ON/OFF edges,
9. frequency analysis using FFT,
10. final classification based on a combination of time-domain and frequency-domain metrics.

3 Signal Detection and Tracking

A YOLO model was used for signal detection. The output of the detection pipeline is a CSV file containing the frame number, bounding box coordinates, detection label, confidence score, and `detection_ID`. For batch processing, it is useful to automatically select the longest track, because the same `detection_ID` does not necessarily correspond to the correct signal in every video.

4 Color and Brightness Analysis

Each extracted crop is normalized to a fixed size. Then, in the HSV color space, the masks `yellow_mask`, `white_mask`, and `active_mask` are created. Based on these masks, the global brightness, brightness of the active region, and a combined brightness signal are computed. The combined signal prioritizes the active region if the mask is sufficiently reliable.

5 Time-Domain and Frequency-Domain Analysis

The brightness signal is detrended using a moving average. This suppresses slow changes caused, for example, by camera exposure, scene motion, or changes in the bounding box size.

Two main methods are used for classification:

1. edge detection in the detrended signal,
2. FFT analysis of the detrended signal.

The monitored frequency bands were:

$$0.60 \leq f \leq 1.25 \text{ Hz}$$

for slow blinking, and

$$1.35 \leq f \leq 2.25 \text{ Hz}$$

for fast blinking.

6 Track Quality Diagnostics

During the work, it became clear that classification of the brightness signal alone is not sufficient. Some errors are not caused by an incorrect decision threshold, but by a poor-quality input track. Therefore, diagnostics of the track and bounding box quality were added.

The following aspects are evaluated in particular:

- track coverage within the analyzed segment,
- number of valid crops,
- ratio of the analyzed part to the full video,
- stability of the bounding box center,
- stability of the bounding box size.

Flags such as `SHORT_TRACK`, `SHORT_ANALYSIS_WINDOW`, `HIGH_BBOX_CENTER_JITTER`, and `UNSTABLE_BBOX_SIZE` are saved into the report. These flags help determine whether the problem is caused by the blink classification itself or already by the input detection.

7 Experimental Results

Testing was performed on two sets of videos:

- `blinking_yellow_signal` – videos with a blinking signal,
- `one_yellow_signal` – videos with a continuously illuminated signal.

Each set contained 17 videos, resulting in a total of 34 videos. After rule calibration and the addition of track quality diagnostics, the following results were achieved:

Metric	Value	Note
Number of videos	34	17 blinking, 17 non-blinking
Correctly classified	28	out of 34 total
Overall accuracy	82.35 %	28/34
True positives	11	correctly detected blinking signals
False negatives	6	blinking signals classified as steady
True negatives	17	correctly detected steady signals
False positives	0	no steady signal was classified as blinking
Precision for blinking	100 %	11/11
Recall for blinking	64.7 %	11/17

Table 1: Summary of results after algorithm calibration.

8 Achievements

The most important result is the elimination of false positive detections in the tested dataset. All videos from the `one_yellow_signal` dataset were correctly classified as continuously illuminated yellow signals. This is practically important because a false positive blinking detection could lead to an incorrect interpretation of the signal state.

The following goals were also achieved:

- creation of a standalone diagnostic script,
- integration of the detection pipeline with the subsequent signal analysis,
- saving of complete diagnostic CSV files,
- generation of brightness signal and FFT plots,
- generation of text reports explaining the decision,
- addition of YOLO track quality metrics,
- significant improvement in accuracy compared to the first version.

The first strict version of the algorithm achieved approximately 50 % accuracy because it classified almost all videos as `steady_yellow`. After rule calibration, the accuracy increased to approximately 82 %.

9 Limitations

The main remaining limitation is the lower recall for blinking signals. Six blinking videos were still classified as `steady_yellow`. The algorithm is therefore conservative: it does not confirm blinking unless the time-domain and frequency-domain evidence is sufficiently strong.

The most problematic cases are those where:

- the dominant FFT frequency is outside the expected band,
- the edge frequency does not agree with the FFT frequency,
- the relative blinking amplitude is very low,
- the track covers only a short part of the video,
- the bounding box jumps significantly,
- the bounding box size is unstable.

For some signals, it was shown that the problem is not in the classification itself, but in the input detection or bounding box stability. If the crop does not consistently follow the same part of the lamp, the brightness analysis may measure changes in the position or size of the crop rather than the actual blinking.

10 Influence of the DETECTION_FPS Setting

The following value was tested:

$$\text{DETECTION_FPS} = 15$$

For a 30 FPS video, this means that YOLO detection is performed approximately on every second frame.

A higher `DETECTION_FPS` may improve track coverage and reduce the number of invalid crops. However, in the measured results, the overall accuracy did not change. This indicates that the remaining errors are not caused only by sparse detection, but rather by bounding box stability or by the nature of the signal itself.

11 Recommendations for Future Work

For further development, we recommend:

1. Add an intermediate state `probable_blinking_yellow`.
2. Stabilize the bounding box over time by filtering its center and size.
3. Improve automatic selection of the correct track.
4. Analyze multiple candidate tracks and select the one with the best mask.
5. Measure brightness in a robust local region inside the bounding box.

6. Evaluate the algorithm on a larger and more diverse dataset.
7. Move the decision rules into a configuration file.
8. Automatically test multiple combinations of `DETECTION_FPS` and `MAX_HOLD_FRAMES`.

12 Conclusion

The proposed algorithm can reliably distinguish a continuously illuminated yellow signal from a portion of blinking yellow signals. The main achieved advantage is the zero number of false positive detections in the tested dataset. If the algorithm classifies a signal as blinking, this result is highly reliable within the tested data.

The disadvantage is that some weak or poorly tracked blinking signals are classified as steady. Further improvement should therefore focus mainly on stabilizing the input bounding box, improving track selection, and introducing an intermediate state for uncertain cases.